Database Management Systems
CPS 3740

Lecture 05

Computer Science, Kean University
Dr. Ching-yu (Austin) Huang
Instructor Paolien Wang

Outline

- Chapter 3: Database Architectures and the Web Transparencies
- Client/Server
- Web DB environment
- Chapter 29: Web, CGI, PHP MySQL
  - Dynamic web page – display data from a database
  - Web DB Programming - PHP MySQL

Chapter 3

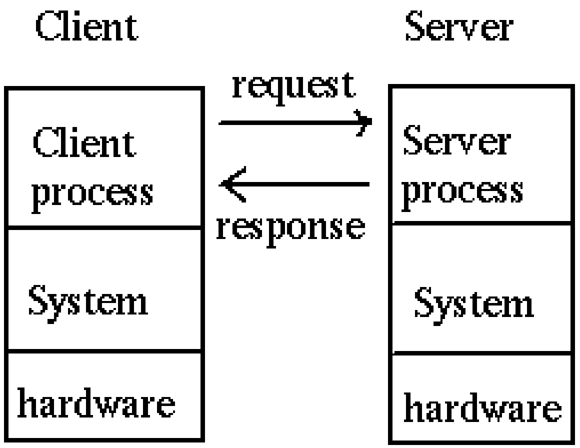
Database Architectures and the Web
Transparencies

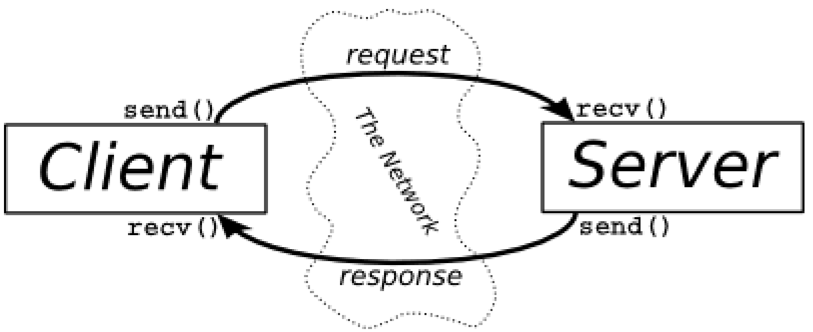
Database Environment - Objectives

- Client–server architecture
- The advantages of this type of architecture for a DBMS.
- Three-tier architectures.

Client /Server

- Client/server is a program relationship in which one program (the client) requests a service or resource from another program (the server).
- Client and server must communicate through certain protocol (port#, process)
- Example:
  - Webserver: google, cnn
  - Client: browser
  - httpd protocol, port 80 (default)
- A machine can be a server and client at the same time.





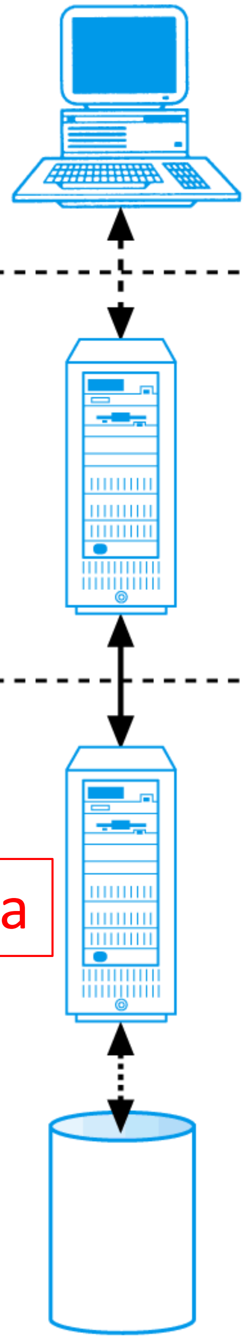
Three-Tier Client-Server

- By 1995, the web browser had more functions and became popular, three layers was proposed - each potentially running on a different platform.
- Advantages:
  - ‘Thin’ client, requiring less expensive hardware.
  - Application maintenance centralized.
  - Easier to modify or replace one tier without affecting others.
  - Separating business logic from database functions makes it easier to implement load balancing.
  - Maps quite naturally to Web environment.
- Our projects are in Three-Tier client-server Architecture.

Middle-tier: Application Server and Web server

- The application server (intermediate layer) is a web server between browser and the database server.
- It runs application programs and set business rules/functions such as login, take orders, etc.
- To avoid users directly access database.
  - **Security**: Database can be inside firewall and application server is in **dem**ilitarized **zone** (DMZ).
  - **Integrity**: Only specific application can access/modify the database

Three-Tier Client-Server



First Tier

Client

Browser

Tasks

- User interface

Second Tier

Application server

Bank website

Tasks

- Business logic
- Data processing logic

Third Tier

Database server

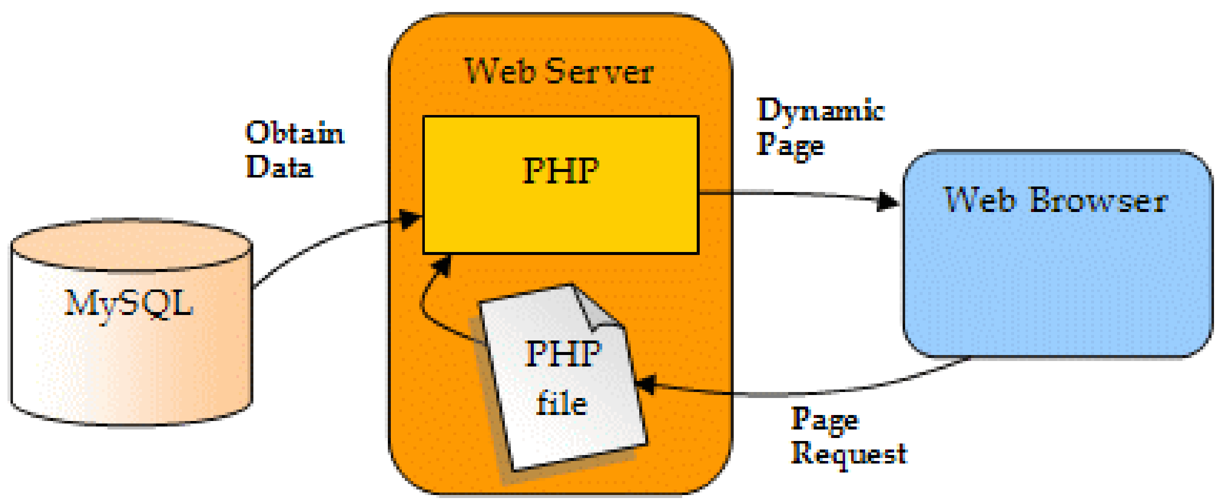
Bank money data

Tasks

- Data va idation
- Database access

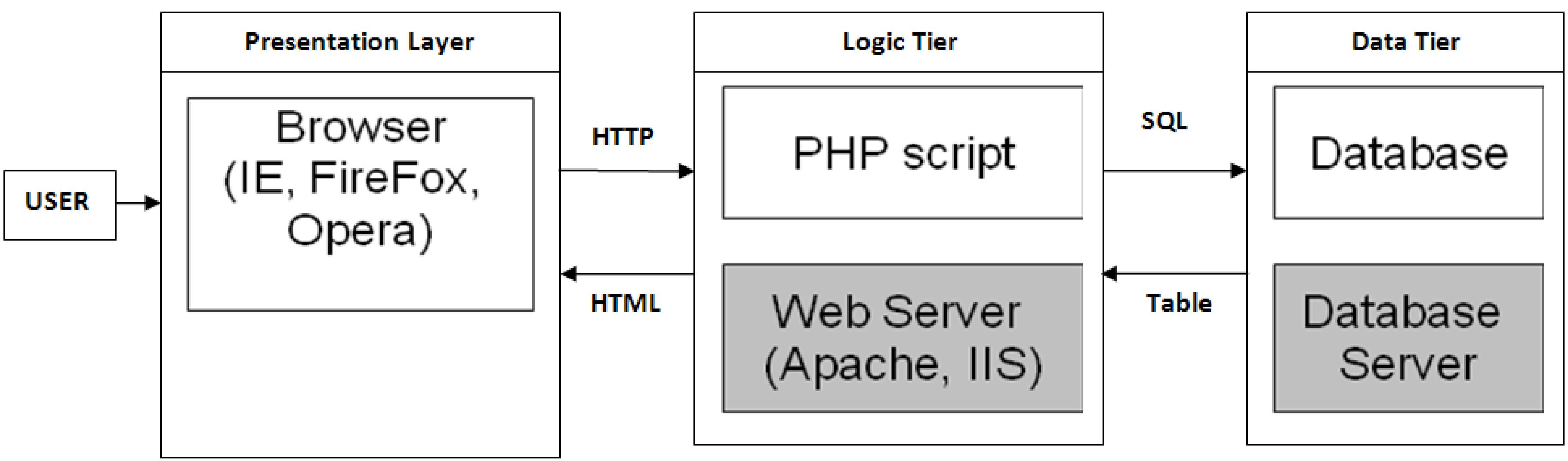
Data in the 3 Tiers Client-Server

- Our Project
  - Tables on the database server
  - PHP, HTML files on web server (application sever)
    - Web pages are on the web server (business)
  - Browser: Your PC



Web technology in 3-tier architecture (our project)

- Front-end (1st tier): browser on your PC
- Middle-end (2nd tier): PHP & HTML files, webserver
- Back-end (3rd tier): tables, database server
- Study this link
    - https://alitarhini.wordpress.com/2011/01/22/concepts-of-three-tier-architecture/



Project - Client vs Server

- Based on process, NOT hardware
  - Required specific protocol/port
  - httpd (80), sshd (22), sftpd (22), mysqld (3306)
- Server (software): provides the service
  - A daemon process is running 24x7 to get/respond request
  - httpd, sshd, mysqld
- Client (software): requests service
  - Brower, putty, filezilla, mysql
- A computer can have both server and client running at the same time
  - The webserver machine responds to the browser's request
  - The webserver machine sends request (mysql is the client) to MySQL server to retrieve data.

Check Daemon Processes on Linux

- See the daemon and process ID

  $ps

- View all running processes information on the system, display the results one screen page at a time using pipe

  $ps –ef | more

- grep specific running processes' names with pipe

  $ps –ef | grep httpd

  $ps –ef | grep sshd

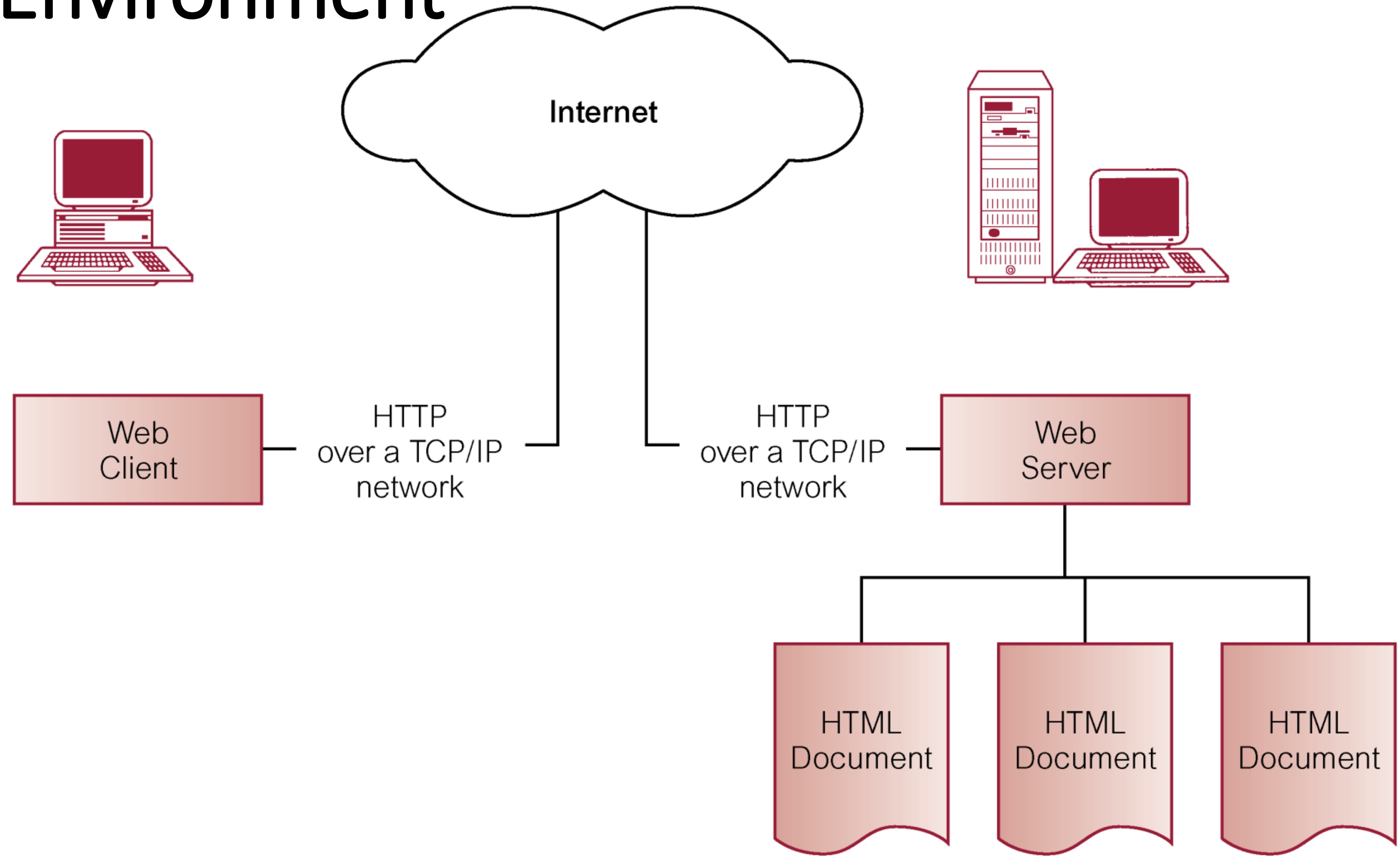
  $ps –ef | grep mysqld

  →what do you see? xxxd means daemon (server)

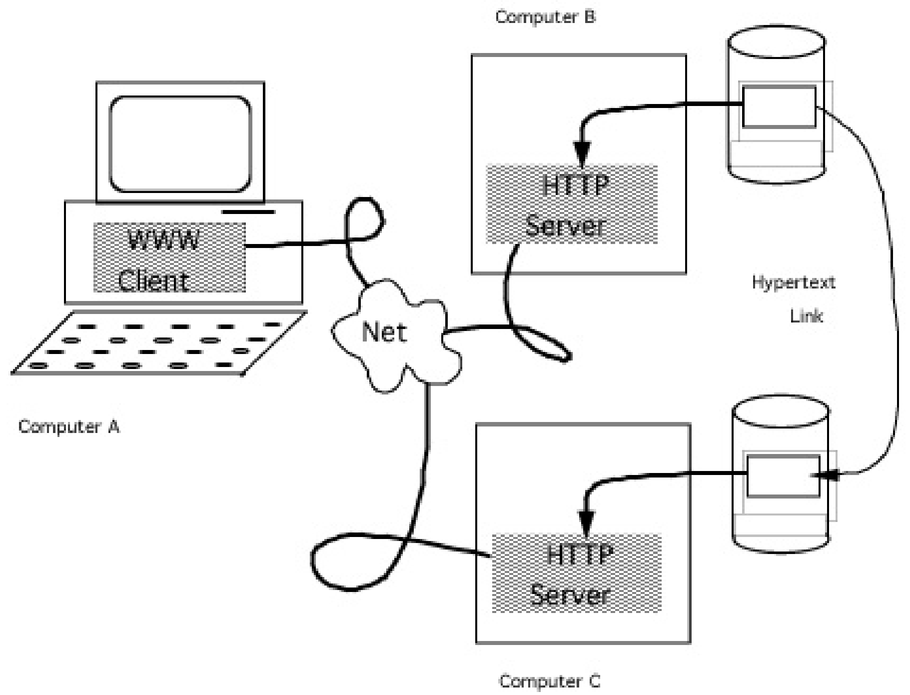
Chapter 29

Web Technology and DBMSs

29.2 Basic Components of Web Environment



29.2 HyperText Transfer Protocol (HTTP)



- Protocol used to transfer Web pages through Internet.
- Based on request-response paradigm:
  - Connection: Client establishes connection with Web server
  - Request: Client sends request to Web server
  - Response: Web server sends response (HTML document) to client
  - Close: closed by Web server

Uniform Resource Locators (URLs)

- Defines uniquely where documents (resources) can be found.
- URL consists the followings:
  - protocol used for the connection (required)
  - host name (required)
  - path name on host where resource stored (required)
- Can optionally specify:
  - port through which connection to host should be made
  - File name, the content sent to browser
  - query string

https://www.w3schools.com/html/

Web Programming

- Front-end programs at Browser
  - Graphic User Interface
  - Java, Javascript, HTML, CSS, Flash
- Middle-end programs at Web server
  - Process, access data from database servers.
  - Usually written in a scripting language, but can be written in any programming language.
  - PHP will be used in our project.
- How does front-end browser pass data to back-end programs?
  - Needs an interface (method) between them.

29.4 Common Gateway Interface (CGI)

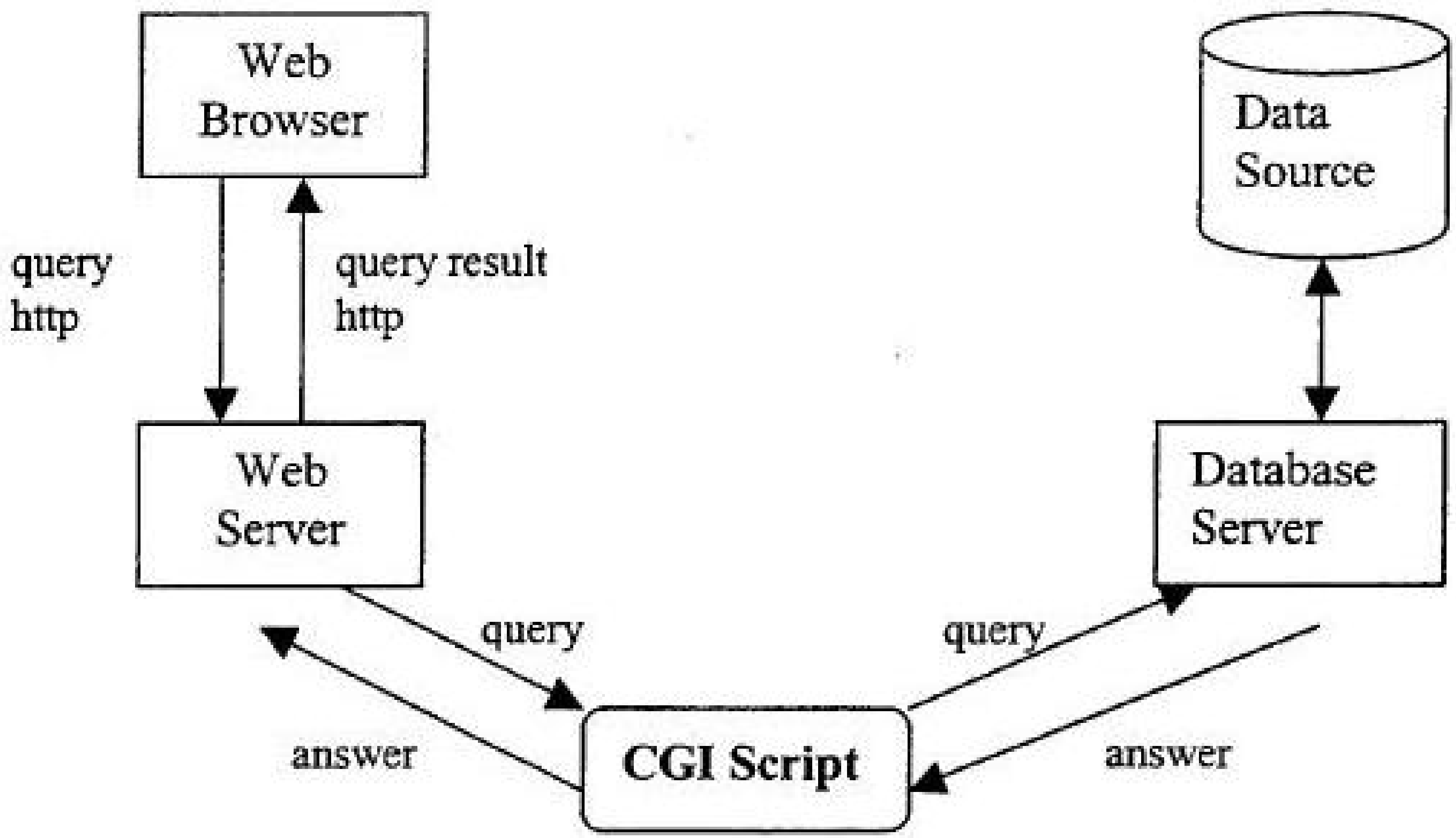
Specification for transferring information between a Web server and a middle-end (CGI) program.

- The web server only intelligent enough to send documents and to tell browser what kind of document it is.
- But the web server also knows how to launch other programs.
- When the web server sees that URL points to a program (script), it executes script and sends back script's output to browser as if it were a file.
- Users can interact with web server through CGI: input, output, access other computers.

About CGI

- CGI defines how scripts communicate with Web servers.
- A CGI script is any script designed to accept and return data that meets to the CGI specification.
- Before server launches script, server prepares number of environment variables representing current state of the server, who is requesting the information, and so on.
- CGI scripts can be written in almost any language, provided it supports reading and writing of an operating system's environment variables.

Database Access on the Web Using CGI Scripts



CGI - Advantages

- CGI is the *de facto* standard for interfacing Web servers with external applications.
- Possibly most commonly used method for interfacing Web applications to data sources.
- Advantages:
  - simplicity,
  - language independence,
  - Web server independence,
  - wide acceptance.

CGI - Disadvantages

- Communication between client and database server must always go through Web server.
- Lack of efficiency and transaction support, and difficulty validating user input inherited from statelessness of HTTP protocol.
- HTTP never intended for long exchanges or interactivity.
- Server has to generate a new process or thread for each CGI script.
- Security.

Class exercises - if and loops in PHP

- Very similar to Java, only difference – variables begin with $

```
<?php
  for ($i=0; $i< 10; $i++) {
       if ($i > 2 && $i <= 5) {
          echo "<br>$i is within the range\n";
        }
    }
?>
```

- Type the above codes, save as “test.php”, and run the program on the command-line
- Put it under your CPS3740 folder on our web server and run it on the browser:

  http://{webserver}/~xxxx/CPS3740/test.php

  - Don’t forget to change its permission mode to 705

Process control – infinite loop

- Check process status:$ ps –ef | grep your_program
- Find the process id
- Kill a process: $ kill -9 process_id
- Ctrl + c
- Ctrl + z
- Resume each suspended job job spec in the background: fg
- All tasks: top

PHP Arrays

Product_id	Name
1	car
2	toy
3	table

- Database query results
  - Two-dimensional arrays
  - First dimension represents rows of a table
  - Second dimension represents columns (attributes) within a row
- Main types of arrays in PHP: $result[]
  - Numeric: index starts at zero, e.g.: $p[0]="car";
  - Associative: pairs of (key => value) elements, e.g.: $p['name']="car";
- PHP **count()** function: count($result)

PHP Array

```
<?php
$array=array(2,4,6,8,10);
$data=array("Feb","Apr","Jun","Aug","Oct");
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
$length = count($array);
for ($i = 0; $i < $length; $i++)
 { print $array[$i] . "->" . $data[$i] . "\n"; }
foreach($age as $x => $x_value)
 { echo "Key= $x, Value=$x_value\n"; }
?>
```

```
$age['Peter'] = "35";
$age['Ben'] = "37";
$age['Joe'] = "43";
```

```
2->Feb
4->April
6->June
8->August
10->October
Key= Peter, Value=35
Key= Ben, Value=37
Key= Joe, Value=43
```

PHP Arrays - Example

- PHP Arrays are dynamic

```
<?php
   $pi[0][0]=1; $pi[0][1]="car";
   $pi[1][0]=2; $pi[1][1]="toy";
   $pi[2][0]=3; $pi[2][1]="table";
   echo "# of row: " . count($pi) . "\n";
   echo "# of col at row 1: " . count($pi[1]) . "\n";
   $pa[0]['Product_id']=1; $pa[0]['Name']="car";
   $pa[1]['Product_id']=2; $pa[1]['Name']="toy";
   $pa[2]['Product_id']=3; $pa[2]['Name']="table";
   echo "# of row: " . count($pa) . "\n";
   echo "# of col at row 1: " . count($pa[1]) . "\n";
?>
```

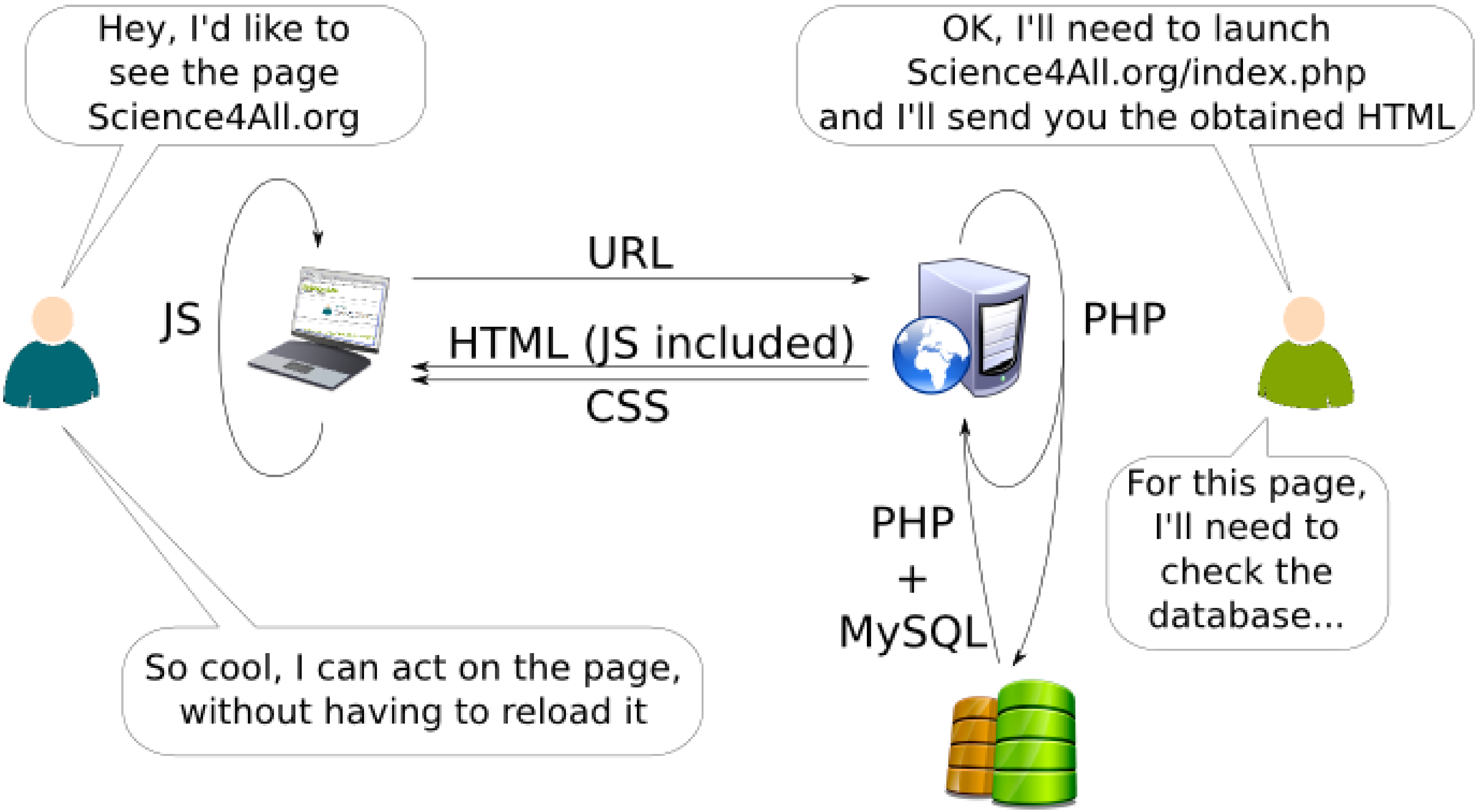
```
+------------+-------+
| Product_id | Name  |
+------------+-------+
|          1 | car   |
|          2 | toy   |
|          3 | table |
+------------+-------+
```

```
php t.php
# of row: 3
# of col at row 1: 2
# of row: 3
# at col at row 1: 2
```

Web database application: How to Display Data?

- Data is stored in database.
- Human cannot visualize data without an interface (text or GUI).
- A web application or web app is any software that runs in a web browser
  - It is created in a browser-supported programming language (such as the combination of JavaScript, HTML and CSS) and relies on a web browser to render the application
- Web application (browser based GUI) is much easier to develop than standalone application.
  - That is why our projects are web based.
- PHP MySQL functions will allow to access data in the database. We will have more discussions later.

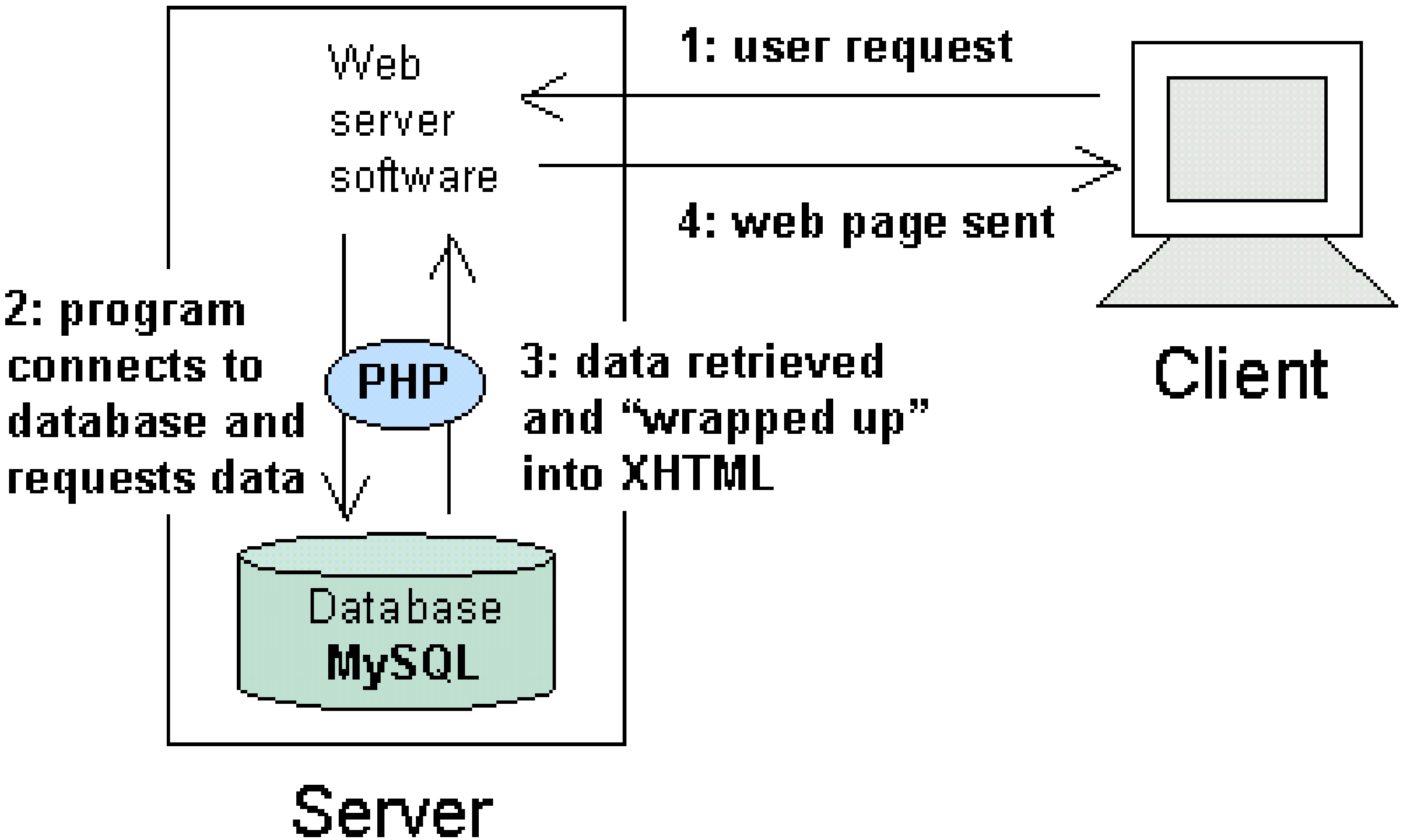
Dynamic web pages with PHP & MySQL



http://www.knowtex.com/nav/languages-of-the-web_37098

PHP, Web, DB Connectivity

- Dynamic pages are often built on the fly after querying a database.



Web Database Programming

- A web database program requires special APIs or libraries to connect to a database server.
- A PHP program requires special MySQL functions to access a MySQL database server.
- PHP MySQL API functions: mysql_*()
  - mysql_connect(), …
  - This extension is deprecated as of PHP 5.5.0
- PHP mysqli class
  - mysqli_connect(), …
  - http://php.net/manual/en/book.mysqli.php
- Cannot mix mysqli_*() and mysql_*() functions

DB Connection Steps

- Open connection
- Select database
- Send and execute query
- Fetch data from database server to result set (buffered at client-side)
- Access every row in result set
- Process the fields in every row
- Free up result set
- Close connection

PHP & MySQL

- Steps for PHP to connect to MySQL:
  - Open a Connection to the MySQL Server

  $con =mysqli_connect($host, $username, $password,$dbname);
  - Execute the SQL query and return records

  $sql = **"SELECT id, name FROM** CPS3740.Customers **";**

  $result = mysqli_query($con, $sql);
  - Fetch the data from the database, one row

  $row = mysqli_fetch_array($result)
  - Free the result set

  mysqli_free_result($result);
  - Close a Connection

  mysqli_close($con);

Define dbconfig.php

- This centralized file is used to define the PHP variables for the login and database information:

```
<?php
$host = "imc.kean.edu"
$username = "XXXX", your database login
$password = "pppp", your database login password
$dbname = "current class database name"
?>
```

- This file will be included in all your PHP programs that need to access the database.

include dbconfig.php

- Put the database variables into dbconfig.php
- All PHP programs accessing database can have:

```
include "dbconfig.php";
```

- Why to have a centralized file?
  - What if you need to change the password?
  - Good for maintenance and security
- It helps:
  - Easy maintenance
  - More protection

Connect to Database

- Connect to database in PHP
- PHP statements

```
$con = mysqli_connect($host, $username, $password, $dbname)
or die("<br>Cannot connect to DB:$dbname on $host\n");
```

- Use die() function to display the message and then terminate the program.
- Do not continue the program if the database connection is failed.
  - Not waste the application server resource
  - Eg. CPU, hard drive, network, etc.

Send and Execute Query in Database

- Database column names in PHP query variable must be the same as database column names.
  $sql= "SELECT **login, password, salary** FROM CPS3740.Customers";

- PHP query statement can contain PHP variables
  $sql= "SELECT login, password FROM CPS3740.Customers where login='**$username**' ";

- Return result set to PHP variable
  $result = mysqli_query($con, $sql);

Free Result Set & Close Connection

- **Free result set. It should be inside the if ($result) statement**
  mysqli_free_result($result);

- **You should have the mysqli_free_result only if the $result is not empty or null. Otherwise, there will be a warning.**
  - **Warning**: mysqli_free_result() expects parameter 1 to be mysqli_result

- **Close database connection**
  mysqli_close($con);

Example:
Use PHP mysqli_ to work with Database

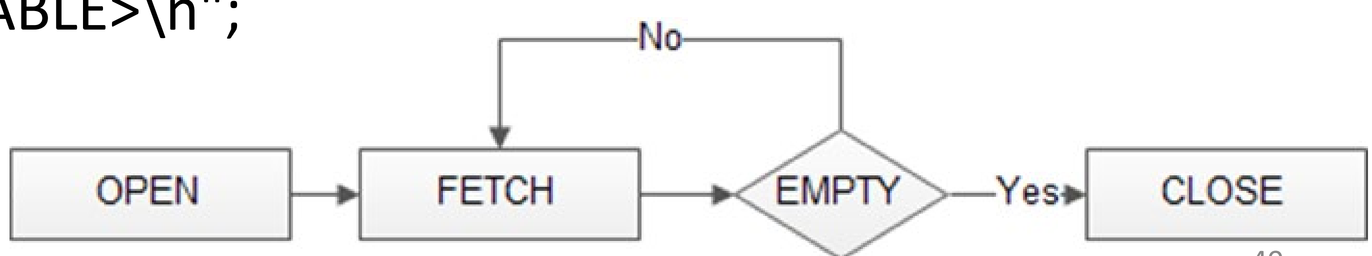
```
<?php
include "dbconfig.php";
$con = mysqli_connect($host,$username, $password, $dbname);

$sql = " SELECT id, name FROM CPS3740.Customers ";
$result = mysqli_query($con, $sql);

if($result) {
    while($row = mysqli_fetch_array($result)){
        $name = $row["name"];
        if ($name <>"")       echo "<br>". $name . "\n";
    }
}
mysqli_free_result($result);
mysqli_close($con);
?>
```

Fetch Data from Database and Format into an HTML Table

- Key name in $row array must be the same in query
- Use mysqli_fetch_array function to read the result set.

```
if($result) {
  if (mysqli_num_rows($result) > 0) {
     echo "The following customers are in the database.";
     echo "<TABLE border=1>\n";
     echo "<TR><TD>Login<TD>Password\n";
     while($row = mysqli_fetch_array($result)){
        $name= $row["name"];
        echo "<TR><TD>$name\n";
     }
      echo "</TABLE>\n";
  }
}
```

Must be the same in the SQL, case sensitive.



Example: Display Customers, part I

```
<?php
 echo "<HTML>\n";
 # keep the sensitive information in a separated PHP file.
 include "dbconfig.php";

 $con=mysqli_connect($server,$login,$password,$dbname)
       or die("<br>Cannot connect to DB\n");

 $query = "SELECT id, login ";
 $query = $query . " FROM CPS3740.Customers ";
 echo "<br>query: $query\n";

$result = mysqli_query($con,$query);
```

Example: Display Customers, part II

```
if($result) {
    if (mysqli_num_rows($result)>0) {
        ……. see part III
    }
    else {
        echo "<br>No records in the database.\n";
        mysqli_free_result($result);
    }
}
else {
    echo "<br>Something wrong with the SQL query.\n";
}
mysqli_close($con);
?>
```

Example: Display Customers, part III

```
echo “<br>The following customers are in the database.";
echo "<TABLE border=1>\n";
echo "<TR><TD>ID<TD>Name\n";

while($row = mysqli_fetch_array($result)){
  $id = $row['id'];
  $name = $row['name'];
  echo "<TR><TD>$id<TD>$name\n";
}

echo "</TABLE>\n";
```

Class exercise - PHP MySQL

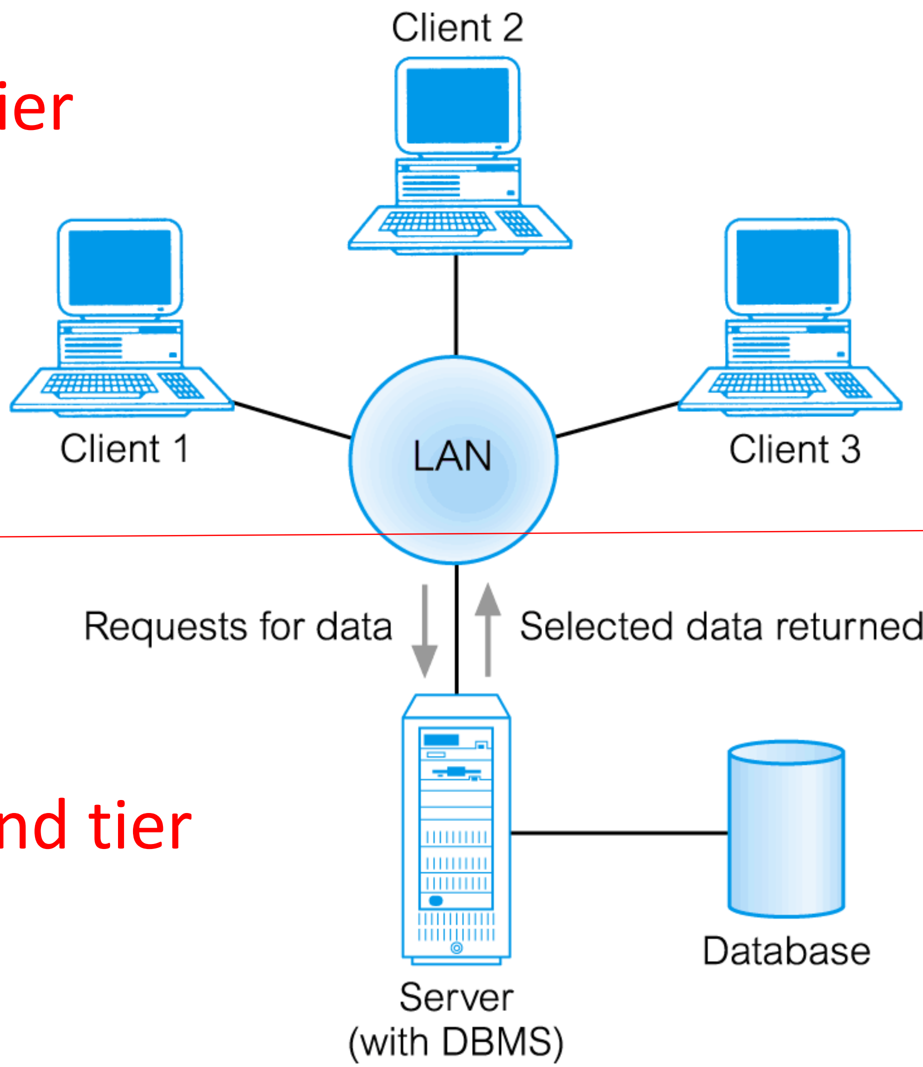
- Make display_customers.php working on the web server
    - http://{webserver}/~xxxx/CPS3740/display_customers.php
    - Display customer id, name in HTML <TABLE>

Error or warning messages

- What will happen and how to handle the following situations:
  - Connection fails – wrong login, password, no permission
    - →Show the error message
    - →Stop the PHP program
  - SQL statement errors – typo, syntax
    - →$result is FALSE (NULL)
    - →Show the error message – "SQL or DB problems"
  - SQL statement returns 0 rows
    - →$result is TRUE
    - →Show a message - "No record found

3.1.3 Traditional Two-Tier Client-Server

First tier



Tasks

- User interface
- Main business and data processing ogic

Second tier

Tasks

- Server-side validation
- Database access

Self study

3.1.3 Traditional Two-Tier Client-Server

- Client (tier 1) manages user interface and runs applications.
- Server (tier 2) holds database and DBMS.
- Advantages include:
  - wider access to existing databases;
  - increased performance;
  - possible reduction in hardware costs;
  - reduction in communication costs;
  - increased consistency.

Self study

Home exercise – PHP programming 1

- Write a PHP program to
  - show the client's IP, OS and browser information.
  - Assign an sorted array (1, 2, 2, 2, 5, 6, 6, 8, 9, 9)
  - Print the sum of the even numbers in the array on the browser. (26)
  - Print the duplicated numbers. (2, 6, 9)
- If you cannot complete this program in 2 hours,
  - do more exercises.
  - get help from Code Samurai ASAP.

Self study

Home exercise – PHP programming 2

- Write a web page that has 2 input boxes that allow users to enter one number in each box.
- Write a PHP program to
  - Use the POST method
  - Receive the 2 values from the browser
  - Print the bigger number and the difference between these two numbers on the browser.
- If you cannot complete this program in 2 hours,
  - do more exercises.
  - get help from Code Samurai ASAP.

Self study

Reading Assignment

- 29.2 The web
  - 29.2 the Web
  - HTML https://www.w3schools.com/html/
- Build your home page on our web server
- 29.3 Scripting Languages – PHP
  https://www.w3schools.com/php
- 29.4 HTML FORMS and CGI
  https://www.w3schools.com/php/php_forms.asp

Self study

PHP, web, database and Linux

- References
  - http://imc.kean.edu/
  - http://mally.stanford.edu/~sr/computing/basic-unix.html
  - https://www.w3schools.com/html
  - https://www.w3schools.com/php/
  - https://www.w3schools.com/php/php_mysql_intro.asp
- Do a lot of exercise
  - Tools – putty (Windows), terminal (Mac), ssh, FileZilla
  - Linux command
  - HTML, PHP, PHP MySQL
- Please work on the project ASAP and go to see Samurai as early as possible if you need any help.

Self study

Conclusion

- Understand client-server
- Understand 3-tier architecture
- Understand CGI
- Practice GET and POST methods
- Start working on Project ASAP
- Practice Linux commands

Self study